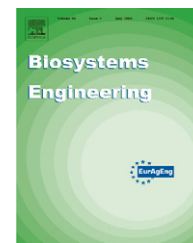


Available at www.sciencedirect.comjournal homepage: www.elsevier.com/locate/issn/15375110**Research Paper: SW—Soil and Water****Tine-influenced infiltration patterns and informing timing of liquid amendment applications using brilliant blue dye tracers**

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Tillage associated with slurry applications, as well as soil physical properties at time of application, imparts strong control over resulting slurry application-induced infiltration patterns and contaminant flow pathways. This field-based study used Brilliant Blue-dyed water applied to tine-influenced silty-clay loam soil, at different antecedent soil water contents, to investigate the nature of infiltration patterns that might be expected to occur during a typical liquid manure or biosolid land application. The tines were associated with a furrow-based Kongskilde Vibro-Flex S-tine slurry application system and a soil pocket-forming AerWay[®] SSD rolling aerator-type slurry application system. For the furrow-based approach, liquid was applied to a small furrow stretch at a nominal rate of 180,000 l ha⁻¹. For the AerWay treatment, a nominal liquid application rate of 180,000 l ha⁻¹ was applied to individual soil pockets. Dye infiltration attributes were determined in 10 cm depth increments to over 1 m beneath the application area. Attributes included: dye infiltration depth, dye spread, and area coverage of dye. The attributes were determined for each depth via image processing software. Experiments were conducted at soil water conditions ranging from 15% v/v to 39% v/v. For furrow-based methods, dye was primarily absorbed above 30 cm depth (above a plough pan which, was approximately 20 cm depth) for all water content conditions. Dye spread:area ratios were generally larger below the pan than above it. Dye penetration depth for all water content conditions exceeded 60 cm depth. With the soil pockets caused by the AerWay treatments, the depth of dye penetration was non-linearly related to the soil water content where greatest dye penetration depth occurred at lower surface soil water contents (< 20% v/v) and at higher ones (> 29% v/v). Overall, for the lower soil water content conditions, desiccation cracks and soil fractures facilitated by tillage action augmented bypass-based preferential flow to continuous sub-pan macropore networks. The greater infiltration depths achieved for the wetter soil conditions were due to greater intrinsic pore sizes that are operative at lower soil tensions and reduced dye sorption. The area covered by dye was greater above the plough pan; but beneath the pan, discrete macropore-based flow (e.g., worm burrows, root channels) dominated infiltration to the deepest observed depths as supported by observations of

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relatively higher dye spread:area ratios. This study underscores the importance of basing liquid amendment timing regulations not only on an upper water content limit but also on a lower limit as well. In addition, the study supports the impact of plough pans augmenting sorption in the tillage layer, thus minimising the degree of preferential flow to depth during a typical land application.

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1. Introduction

Human health risks resulting from liquid manure/biosolid application on agricultural land have gained revitalised interest in the past few years in Ontario, Canada (O'Connor, 2002). Characterisation of the nature of contaminant flow pathways in soils that receive liquid manure or sewage sludge amendments is required to identify application practices that best reduce groundwater and tile drainage water contamination by associated soluble (e.g., nitrates) and non-soluble (e.g., bacteria, parasites) pollutants. Liquid manure/sewage sludge application practices include primarily the following: surface broadcasting/spreading, surface broadcasting followed by incorporation of the amendment by tillage, subsurface or furrow-based injection, and local broadcast over, or injection into, soils immediately tilled (e.g., Wall *et al.*, 1998; Shipitalo & Gibbs, 2000; Bittman *et al.*, 2005; Ball Coelho *et al.*, 2007).

Tillage action, often used during land application practices, is well known to: (i) destroy or disrupt continuous in-situ macropore networks (e.g., large pores in soil such as cracks, burrows, and abandoned root channels) which can otherwise facilitate rapid gravity-based transport of slurry to groundwater/tile drainage systems; and (ii) augment liquid sorptivity in the tillage layers (Azooz & Arshad, 1996; Bandaranayake *et al.*, 1998; Osunbitan *et al.*, 2004; Wahl *et al.*, 2004). Nevertheless, the efficiency by which these tillage practices impart an effect on slurry movement to important water sources and drainage networks will depend, in part, on the soil physical conditions at the time of land application: i.e., soil water content, plasticity, aggregate stability, macroporosity, and associated soil hydraulic flow parameters. Recent work by Turpin *et al.* (2007) indicated that field-saturated hydraulic conductivity of tilled and un-tilled soil will vary with respect to the water status of the soil. For the cultivation layer in silty-clay loam soils, infiltration was shown generally to increase with a decrease in soil water content. Changes in infiltration associated with immediate tine modification of soil in the tillage zone were documented for these conditions; however, it was not determined specifically where flow was occurring in the vadose zone (e.g., lateral, vertical). Such information is important for defining optimal soil physical conditions for applying liquid manures or biosolids to fields to minimise tile drainage and groundwater contamination potential. Modern slurry application methods often utilise an immediate pre-tillage to facilitate direct injection or subsurface deposition, but the nature and changes of amendment flow induced by tillage action under different antecedent soil conditions have not been intensively studied.

For many of the finer-textured soils, soil cracks and tillage-induced soil fracturing during dry soil conditions (Rajaram &

Erbach, 1997) can facilitate gravity flow to depth that effectively bypasses the soil matrix (Prendergast, 1995; Matthews *et al.*, 2000). On the other hand, when soils are wetter, matrix sorptivity is reduced and the larger intrinsic pore network engages flow, which can augment more defined 'macropore'-based flow to depth in the soil profile (Watson & Luxmoore, 1986; Steenhuis *et al.*, 1994). Both situations can pose problems with respect to tile drainage and shallow groundwater contamination via liquid manure/biosolid application (e.g., Wall *et al.*, 1998; Akhand *et al.*, 2006). Hence, knowledge of liquid infiltration patterns (e.g., Kulli *et al.*, 2003; Weiler & Flühler, 2004) associated with tine-soil interaction zones for these contrasting flow conditions may provide a means to identify appropriate times to apply liquid amendments to field soils.

Dye tracers, such as Brilliant Blue FCF, are frequently used to investigate water flow through soil (Andreini & Steenhuis, 1990; Petersen *et al.*, 2001; Reichenberger *et al.*, 2002; Weiler & Flühler, 2004). Brilliant Blue FCF sorbs on soil materials, and German-Heins and Flury (2000) indicate that in terms of toxicity, mobility and visibility, Brilliant Blue is one of the best dye tracers to study water flow processes in the vadose zone.

This study used brilliant blue dye to investigate, in a qualitative manner, the nature of infiltration associated with immediate shallow-tine action via Kongskilde Vibro-Flex S-tine and AerWay[®] SSD rolling aerator-type tines, for many different water conditions of a silty-clay loam soil. These tillage implements, although each distinctively different from each other in terms of their tilling action, are used on shallow slurry injection/deposition systems in Canada (Ball Coelho *et al.*, 2007; Bittman *et al.*, 2005).

2. Materials and methods

2.1. Field site

The study site is located on a tile-drained field located near Winchester, Ontario, Canada (lat. 45°03'N, long. 75°21'W). The soil is classified as a North Gower clay loam (Orthic Humic Gleysol (Canadian), Mollic Gleysol (FAO)). Average $\pm$ standard deviation of soil particle size (hydrometer method) over the study field (Turpin *et al.*, 2007) for 0–15 cm ($N=46$) and 30–50 cm ($N=46$) depth are $(0.21 \pm 0.025 \text{ g g}^{-1} \text{ sand, } 0.38 \pm 0.037 \text{ g g}^{-1} \text{ clay and } 0.41 \pm 0.037 \text{ g g}^{-1} \text{ silt})$, and $(0.10 \pm 0.021 \text{ g g}^{-1} \text{ sand, } 0.42 \pm 0.033 \text{ g g}^{-1} \text{ clay, and } 0.48 \pm 0.034 \text{ g g}^{-1} \text{ and silt})$, respectively. Additional soil survey information on the North Gower soils can be found in Schut and Wilson (1987). The field had been under timothy and brome grass for 9 years prior to 1996 and was never tilled while under hay production.

In 1996, the field was subdivided into crop plots, 300 m long and 13 m wide. From 1996 to 2002, the field was under livestock (maize–maize–alfalfa–alfalfa) and cash (maize–maize–wheat–soybean) cropping management practices. From 1996 to 2002, all plots were conventionally tilled, consisting of autumn mouldboard ploughing and spring cultivation (S-tine type) prior to seeding.

In 2003, the field was in wheat production (Turpin *et al.*, 2007). The field was not cultivated after wheat harvest in August 2003. The experiments were conducted in 2004, and the soil was not tilled and no crops were planted in that year. Herbicides were applied to the field in June and September 2004 and periodic mowing was employed to reduce weed growth. Precipitation data during the study period were obtained from a meteorological station at the site.

2.2. Tillage implements

Two contrasting tillage units with liquid slurry injection systems were used in this study (Fig. 1). These were the Kongskilde (K) Vibro-Flex 2011 (Kongskilde, Strathroy, ON) and a pull-type AerWay[®] SSD (A) (Holland Equipment Limited, Norwich, ON). For the Kongskilde system, disc-hillers and coulters were not employed. The unit was composed of two rows of tines with mulching shares (0.11 m width). Tine spacing on the unit is 28 cm, for a total of 11 tines and with a working width of approximately 3 m. The AerWay[®] SSD slurry applicator was a pull-type with a working width of approximately 3 m with 16 aerator-type tines; each tine with length 20 cm. The angle of the AerWay tines relative to pull direction can be adjusted to modify the degree of soil disturbance and the size of the ‘soil pocket’; the greater the angle, the greater the pocket size and the soil disturbance. Tine angle relative to pull direction was set to 10°; which is the maximum angle setting.

For this study, the nominal tillage depth for both tillage implements was 13 cm; depth tests off the site in similar soil

types were conducted prior to the experimental runs to adjust depth control mechanisms and ballast (the latter for (AerWay)). Ballast was required for AerWay during the drier soil conditions to ensure adequate depth penetration by the tines. Both injector units were operated without liquid application to facilitate subsequent soil physical measurements along tine–soil interface locales. Implement ‘runs’ were performed through the summer and autumn of 2004. Each run represented a different surface soil water content condition (via natural rainfall and drying events) and location in the field. Each tillage ‘run’ includes a Kongskilde and an AerWay tillage, except for runs 6 and 10 (only (AerWay)) and runs 7 and 8 (only Kongskilde). Each tillage run was the width of the implement and 15 m in length (Turpin *et al.*, 2007).

2.3. Blue dye approach

Due to logistical and timing constraints, a maximum of two replicate dye assessments were conducted per tillage treatment for selected runs. For each assessment, 500 ml of Brilliant Blue FCF (C.I. 42090) (5 g l⁻¹) was poured gently into a confined area of soil disturbed by the implement tines, i.e., inside a soil pocket for the AerWay and in a 10 cm furrow stretch for the Kongskilde. The amount of dye corresponded to an application rate of approximately 180,000 l ha⁻¹. Although this rate is high, it is within permitted liquid sewage application rates in Ontario (M. Payne, personal communication). This rate was used to maximise evaluation of dye-infiltration patterns in the entire soil pocket for the AerWay treatment, and furrow side-walls for Kongskilde. Before adding the dye liquid, loose soil in the soil pocket and furrow was cleared away (Turpin *et al.*, 2007). This activity reduced interpretation problems associated with flow pattern comparison between the different ‘runs’; since furrow/soil pocket soil backfill has enormous spatial variability. For the Kongskilde furrow, two metal plates (30 cm × 30 cm) spaced

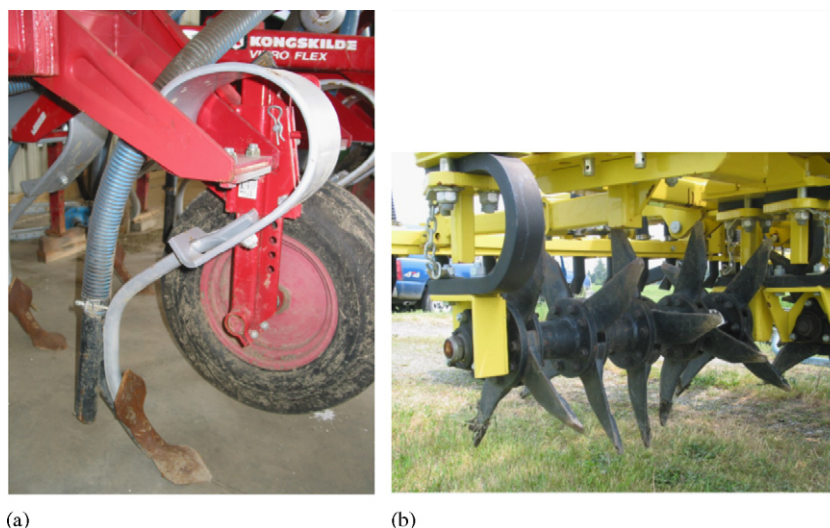


Fig. 1 – Two types of slurry application tillage tines used in this study: (a) Kongskilde Vibro-Flex tine with mulching share. Shown with slurry drop tube attachment and (b) AerWay[®] SSD aerator-type tines.

10 cm apart were inserted vertically (and perpendicular to the furrow) to a depth of 5 cm below the bottom of the furrow to prevent the applied liquid from flowing along the furrow. Short-circuit flow between the plate and soil was minimised by tamping the soil along the plate–soil contact area with an eraser on a pencil. Preparation of the soil pocket for the AerWay experiments consisted of carefully removing the loose soil from the soil pocket and applying dyed liquid directly in the pocket. For both methods, the stained soil was covered with a large tarpaulin (10 m × 10 m) for a 2-day period. A 1 m × 1 m area centred on the dye application zone was manually excavated to depths of 5, 10, 15, 20, 30, 40, 50 cm and at deeper 10 cm increment depths pending infiltration depth findings. At each excavated depth for each measurement site, loose soil was removed using a vacuum and a small brush.

2.4. Soil, digital image, and statistical analysis

Soil water content measurements were made using a TRASE model 6050 TDR (time domain reflectometry) unit using 15 cm three-pronged wave guides for 0–15 cm, 13–28 cm, and periodically at 28–43 cm depth increments. For the Kongskilde treatment, the 13–28 cm depth increment measures, reflected the soil water content immediately below the furrow bottom, and was used for the Kongskilde measurements in this study as water content reference criteria for comparative purposes. While for the AerWay treatment, the 0–15 cm depth TDR measurements served as a soil water content reference. Other soil physical information corresponding to the site and the specific runs are presented in [Turpin et al. \(2007\)](#).

Digital images were taken of each excavated soil depth using a digital camera mounted directly over the measurement area.

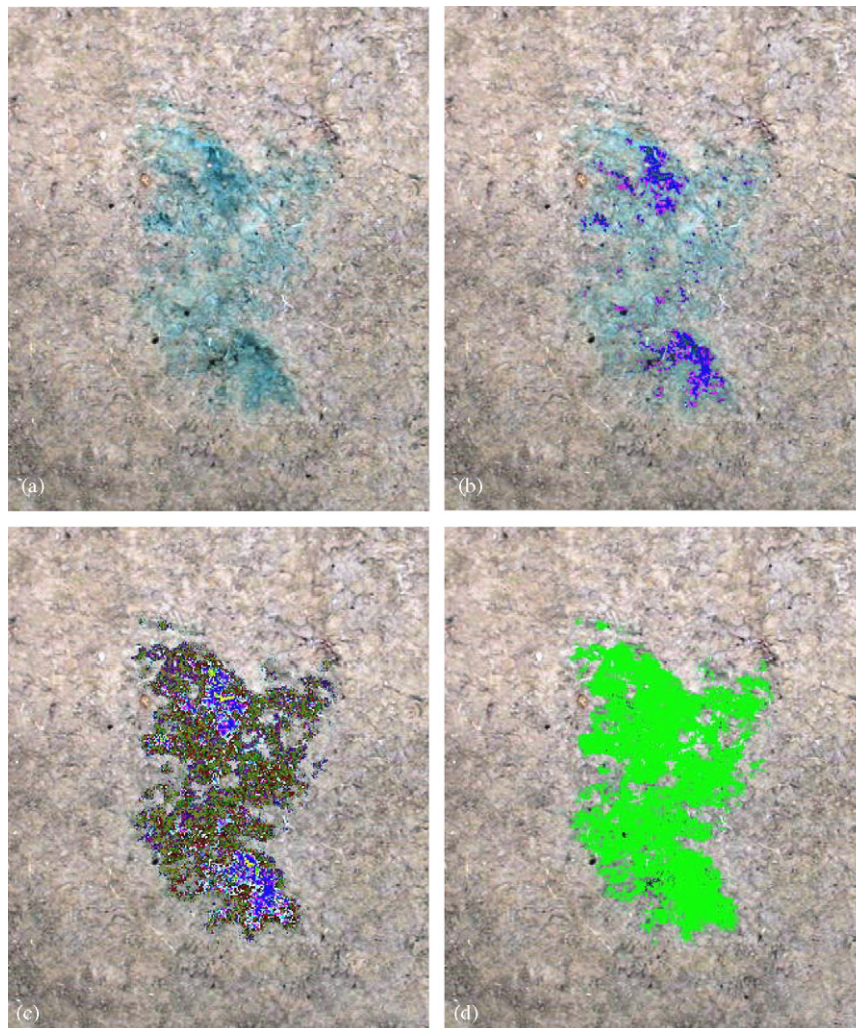


Fig. 2 – Image analysis approach: (a) soil stained by blue dye after a 2-day waiting period; (b) two colour thresholds were defined, and the pixels having RGB values within the colour threshold categories were tagged with blue and pink pixel colours, respectively; (c) thirteen different colour thresholds (represented by 13 different colours) were required to select all the different intensities of blue dye in the stained area; (d) the selected pixels meeting the colour threshold definitions are coloured in green in order to calculate the area coverage of the dye. Note: dye intensity was not singularly differentiated for subsequent analyses.

The digital pictures were analysed using *analysIS* (Soft Imaging System, Lakewood, CO, USA) image-processing software. Images were scaled using an object of known length set in each image setting. Pixels illustrating stained soil were selected by setting thresholds for red, green, and blue (RGB) values. About 13 thresholds were necessary in order to select all the area stained by the dye. The thresholds were expected to change slightly from picture to picture because of variation in soil colour and light intensity. Selected pixels were colour coded in green (RGB = 0, 255, 0), and subsequently, software analyses were performed in order to determine the area (cm²) covered by green-coded pixels (Fig. 2).

Several attributes were measured for each excavated soil layer via the image processing: total dyed area (A), spread of dyed area (S) (variables determined according to Fig. 3), and depth of dye penetration in soil profile (D). Regression tree analysis via the Classification and Regression Tree Analysis (CART) method (Breiman *et al.*, 1984; Steinberg & Colla, 1995) was used in an exploratory manner to uncover dye pattern parameter (A, S, and D) vs. soil water content relationships. CART is a machine-driven, non-parametric, binary, recursive data partitioning approach that uses variance minimising criteria to group-dependent variable data on the basis of independent variable threshold criteria. The CART-based regression tree is a form of decision tree where all dependent

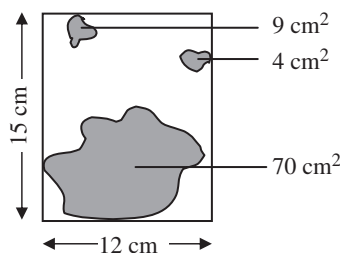


Fig. 3 – Approach to assess, for specific soil layers, total stained area (A) and degree of dye spreading (S). For example in this figure, A = 83 cm² and S = 180 cm² (15 cm × 12 cm).

data are divided, in a manner discussed, into nodal groups; each resulting nodal group can be further subdivided into additional nodal groupings until user-specified minimum node size constraints stop tree development. Nodes that are not further subdivided are called terminal nodes. In this study, the dependent variables were S, D, and A and the independent variable was soil water content. The analysis was used to determine if threshold soil water contents could be used to group-distinctive dye infiltration pattern parameters. The default CART settings and approach to analysis employed are described in *Lapen *et al.* (2001)*.

3. Results

Soil water content, field-saturated hydraulic conductivity and bulk density for each Kongskilde and AerWay run are given in *Tables 1 and 2*, respectively. These measurements were made at sites representing each run according to *Turpin *et al.*, (2007)*. For Kongskilde treatment, the range in average soil water content at the 13–28 cm depth increment was approximately 10% v/v, albeit the minimum and maximum observed water contents were from 15%v/v (run 11) to 34%v/v (run 8), respectively. For the AerWay treatment, the range in average soil water content (0–15 cm depth increment) was approximately 18%v/v. The observed minimum and maximum water contents were 15% (run 10 and 11) to 39%v/v (run 2), respectively. Average field-saturated hydraulic conductivity (log (cm s⁻¹)) for the Kongskilde treatment ranged for –1.68 for run 11 (lowest soil water content run) to –2.65 for run 7 (high soil water content run). For the AerWay treatment, average field-saturated hydraulic conductivity (log (cm s⁻¹)) ranged from –1.77 for run 5 (low soil water content run) to –2.56 for run 6 (highest soil water content run).

3.1. Kongskilde

Depth of infiltration of the blue-dyed liquid for each soil site replicate and run is given in *Fig. 4*. Except for replicate number 2 for run 8, the blue dye reached a depth of at least

Table 1 – Descriptive statistics of logged field-saturated hydraulic conductivity (K_{fs}) determined with pressure infiltrometers in tine-disturbed furrow bottom (infiltration rings inserted at 13 cm depth (*Turpin *et al.*, 2007*)), volumetric soil water content (undisturbed by tine) determined using time-domain reflectometry (TDR) (13–28 cm depth in bottom of furrow), and bulk density (13–21 cm depth) for the Kongskilde treatment

Run	K_{fs} , Log(cm s ⁻¹)					Soil water content, % v/v					Bulk density, kg m ⁻³				
	Mean	SEM	Min.	Max.	N	Mean	SEM	Min.	Max.	N	Mean	SEM	Min.	Max.	N
5	–2.08	0.129	–3.22	–1.32	16	28.1	0.4	22.5	32.3	48	1370	10	1290	1430	16
7	–2.65	0.133	–3.46	–1.89	16	30.2	0.3	23.4	34.2	48	1460	20	1200	1530	16
8	–2.57	0.215	–3.35	–1.93	7	30.9	0.5	27.5	34.4	15	1370	20	1300	1430	7
9	–2.26	0.078	–2.95	–1.80	16	25.4	0.3	20.2	29.5	48	1490	20	1280	1610	16
11	–1.68	0.149	–2.76	–0.94	11	21.1	0.7	15.4	26.4	18	1380	40	1290	1490	5
12	–1.97	0.193	–3.04	–1.19	8	28.9	0.3	26.6	33.1	24	1400	20	1340	1460	8

Note: wilting point of the soil is approximately 11% v/v for this soil.
SEM, standard error of mean.
N, number of samples.

Table 2 – Descriptive statistics of logged field-saturated hydraulic conductivity (K_{fs}) of soil pockets determined with the Guelph permeameter (head set to approximately soil pocket depth (12 cm) (Turpin *et al.*, 2007)), volumetric soil water content (undisturbed by tine) determined using time domain reflectometry (TDR) (0–15 cm depth) and bulk density (0–8 cm depth) for the AerWay treatment

Run	K_{fs} , Log(cm s^{-1})					Soil water content, % v/v					Bulk density, kg m^{-3}				
	Mean	SEM	Min.	Max.	N	Mean	SEM	Min.	Max.	N	Mean	SEM	Min.	Max.	N
2	−2.34	0.114	−3.06	−1.69	16	34.8	0.3	29.6	38.5	48	1400	20	1340	1550	16
3	−2.09	0.092	−2.80	−1.46	15	28.4	0.3	23.7	33.3	48	1390	20	1220	1500	16
4	−1.91	0.094	−2.67	−1.51	16	28.7	0.4	21.5	32.9	48	1440	20	1340	1560	16
5	−1.77	0.101	−2.58	−1.38	15	21.7	0.5	15.5	29.0	48	1360	20	1140	1540	16
6	−2.56	0.153	−4.16	−1.81	14	36.1	0.2	33.5	37.7	48	1380	20	1150	1480	16
9	−1.88	0.145	−3.56	−1.37	16	22.2	0.5	16.5	27.9	36	1450	20	1300	1590	16
10	−1.89	0.155	−3.23	−1.23	16	22.2	0.4	15.3	29.6	45	1340	20	1110	1510	16
11	−1.97	0.209	−3.15	−1.45	8	17.7	0.5	15.3	22.5	16	1350	30	1250	1530	8
12	−2.10	0.178	−3.00	−1.27	12	31.5	0.4	20.9	34.9	36	1370	10	1270	1450	12

Note: wilting point of the soil is approximately 11% v/v.

SEM, standard error of mean

N, number of samples

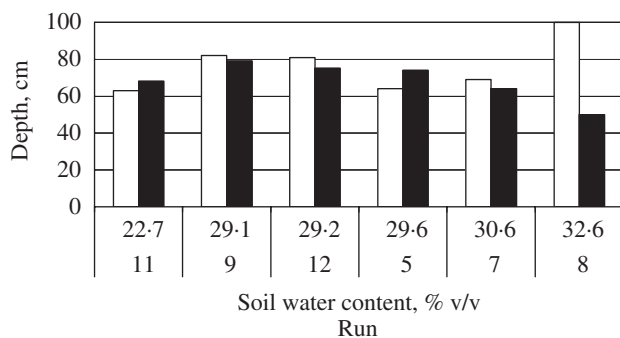


Fig. 4 – Maximum observed depth of dye infiltration for the Kongskilde treatments for specific runs. The soil water contents represented for each run represent the average of six measurements taken immediately around the dye measurement areas: □, replicate 1; ■, replicate 2.

60 cm in all cases (Note: replicate 1 for run 8 exceeded 1 m depth). The large effective application rate was deemed to be at least partially responsible for such uniform (in terms of water content condition) trends. The Pearson's correlation coefficient (r) between observed infiltration depth and run average soil water content (13–28 cm depth) was 0.44. Fig. 5 presents the area (A) covered by the dye and the spreading (S) of the dye for each Kongskilde site and depth excavated. For tractability purposes, the S and A attributes for Kongskilde at 5 and 10 cm depths reflected images taken above the open furrow (13 cm deep). The area and spread variables recorded for these depths reflected exposed dye patterns to approximately 13 cm (bottom of the furrow). There was some indication of a positive relationship between S and soil water content (13–28 cm depth increment) for image-processing depths less than 20 cm, a depth interval within the historical conventional tillage zone in the field. Pearson correlation coefficients between average run S value vs. average run soil water contents (13–28 cm depth) ranged from 0.46 to 0.64 for the 5 to 20 cm depth interval data sets. For depth intervals

below 20 cm, respective correlation coefficients ranged between −0.09 and 0.21. Conventional tillage at the site had been fall mouldboard ploughing to a maximum depth of approximately 20 cm followed by spring cultivation to a depth of approximately 10 cm. The dye spread was greatest at the 50 cm depth interval, relatively, in both the runs with lowest and highest observed soil water contents. For dye soil area coverage, there was an expected decrease with depth for specific runs with a majority of the dye coverage occurring above 20 cm depth (i.e., lateral infiltration above the plough pan layer). Below 20 cm depth, area coverage of dye was significantly reduced for all soil water conditions. For A measures for depth intervals ≤ 20 cm depth, the correlations between average run soil water content and run average A, ranged from 0.22 to 0.74; below the 20 cm depth interval, the correlation coefficients ranged from 0.03 to 0.31. The generally larger S:A ratios below approximately 20 cm depth supports visual observations of discrete macropore flow via worm burrows and small desiccation cracks beneath the plough pan. For the Kongskilde treatment, CART analysis was not employed given the limited data.

3.2. AerWay

The average volumes of the AerWay pockets are given in Table 3 for each run. There was greater general fracturing and clod lifting during the drier conditions, which increased the tillage impacted area to various degrees, and subsequently soil pocket size (run 11 and 5 had highest average pocket volumes) (Turpin *et al.*, 2007). Fig. 6 shows the blue dye infiltration depths for each AerWay run. The greatest infiltration depth observations (i.e., > 60 cm) occurred for runs with average soil water contents (0–15 cm depth) $\geq 31\%$ and $\leq 19\%$ v/v. For run average water contents $\geq 23.7\%$ v/v, there was a positive relationship ($r = 0.88$) between average run infiltration depth and run average surface soil water content (0–15 cm depth). Whereas, for runs with average surface soil

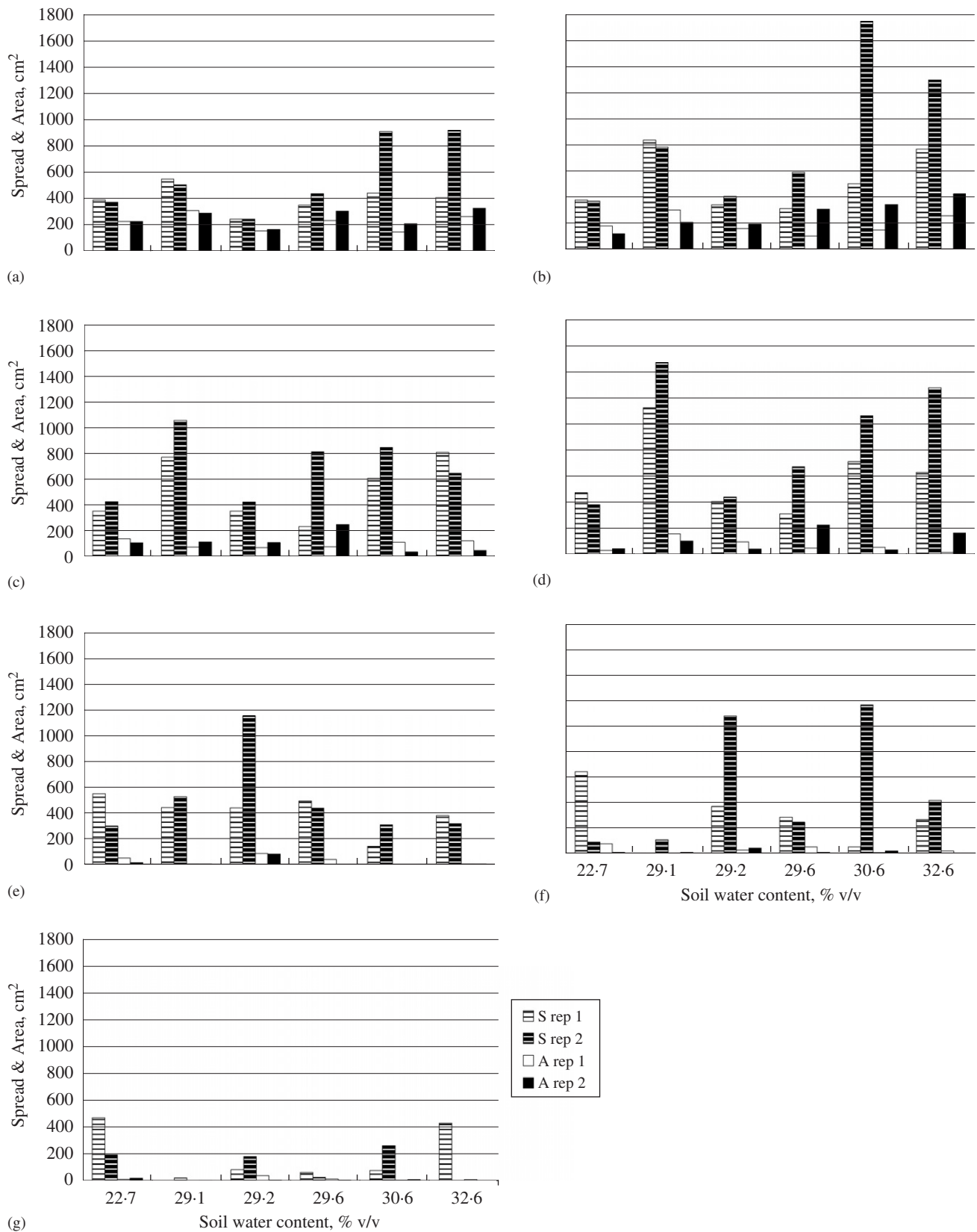
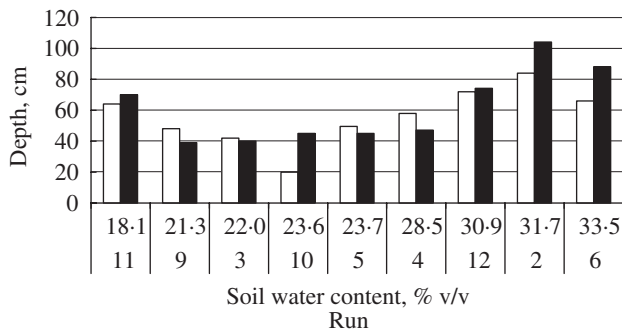


Fig. 5 – Degree of dye spread (S) and dye area (A) coverage for specific soil depths down to 50 cm for Kongskilde treatments: (a) 5 cm depth, (b) 10 cm depth, (c) 15 cm depth, (d) 20 cm depth, (e) 30 cm depth, (f) 40 cm depth, (g) 50 cm depth.

Table 3 – Mean soil volume and standard error of mean (SEM) for each run of the AerWay treatment

Run	Soil volume, cm ³	
	Mean	SEM
1	436	30
2	393	22
3	498	29
4	418	31
5	565	43
6	448	33
9	361	15
10	435	30
11	523	28
12	448	35

**Fig. 6 – Maximum observed depth of dye infiltration for the AerWay treatments for specific runs. The soil water contents for each run represent the average of six measurements taken immediately around the dye measurement areas: □, replicate ; ■, replicate 2.**

water contents $\leq 23.6\%$ v/v, there was a negative relationship ($r = -0.99$) between soil water content and run average infiltration depth. The former relationship was due in part to: (i) reduced soil sorptive capacity with increasing water content; and (ii) the greater soil pore sizes that operate at the higher water content values. On the other hand, the negative relationship between infiltration depth and surface soil water content for average water content values below approximately 24% v/v was attributed to greater desiccation crack-based bypass flow to depth, tillage fracturing, and subsequent exposure of large pores, and potentially a greater degree of hydrophobicity. A simple polynomial regression (coefficient of determination $R^2 = 0.66$) was used to formalise the correlation structure of the data shown in Fig. 6 (for the observed water content range). The regression has the form

$$D = 0.5511\theta_v^2 - 26.429\theta_v + 359.300, \quad (1)$$

where D is the depth of dye infiltration in cm and θ_v is the soil water content in % v/v.

The dye spread patterns for the AerWay soil pocket application are given in Fig. 7. There were either no correlations or very modest positive relationships between run average soil water contents and run average S for soil

depths above 20 cm depth (Pearson correlation coefficients ranged from -0.12 to 0.55). The dye spread vs. soil water content relationships ≥ 30 cm depth mimicked to a large degree the dye penetration depth data in Fig. 6. The correlation coefficients (r) between the average run soil water content and average run S for these depth increments ranged between -0.08 and 0.57 . For dye area coverage, general decreases occurred with depth, with significant decreases for all runs occurring immediately below 20 cm depth, similar to that observed for the Kongskilde treatment. The dye area trends below this depth again follow, for the most part, infiltration depth trends. Pearson correlations between run average A for depths ≤ 20 cm and for depths > 20 cm ranged between -0.30 to 0.11 , and -0.51 to 0.39 , respectively. Like the Kongskilde treatment, the $S:A$ ratios for the AerWay were observed to be relatively higher at deeper depths as likened to macropore flow via discrete flow channels and pathways.

Table 4 summarises the results obtained from the CART-based regression tree analyses. All presented data were cross-validated in CART using a 10-fold (10% testing and 90% learning) cross-validation procedure (Steinberg & Colla, 1995). The regression trees, for a given dependent variable, with the 'best' average accuracy as determined internally in CART via this cross-validation procedure, were selected for discussion here. Dependent variables not presented in Table 4 could not be internally cross-validated in CART.

For the infiltration depth variable (D), the depth data were split by CART into three primary run average soil water content nodal groupings: (i) data for run average surface soil water contents (0–15 cm depth) $\leq 19.7\%$ v/v (average infiltration depth = 67 cm), (ii) data for soil water contents between 19.7 and $\leq 29.7\%$ v/v (average infiltration depth = 43 cm), and (iii) data for soil water contents $> 29.7\%$ v/v (average infiltration depth = 81 cm). Below the plough pan at 30 cm depth, the $\leq 19.7\%$ water content threshold delineated greatest average spread (average $S = 1027 \text{ cm}^2$) relative to, on average, the other water content data groupings. However, the CART-based analysis of S for 20 and 40 cm depths was greatest for surface soil water contents between roughly 30% and 33% v/v. For dye area coverage analyses, results derived for A30 (30 cm depth) and A40 (40 cm depth) were consistent in that they grouped data on the basis of $> 19.7\%$ v/v (≤ 121.6 and 9.3 cm^2 for A30 and A40, respectively), $19.7\text{--}\leq 29.7\%$ v/v (3.7 and 1.2 cm^2 , respectively) and $> 29.7\%$ v/v (13.8 and 16.3 cm^2 , respectively) nodal soil water content groupings where the area of dye coverage was relatively greater for both the lower and higher groupings. For the shallower depths (5, 10, and 20 cm), run average A values vs. average run water content group relationships indicated higher average A for soil water contents between 21.7% and 32.6% v/v, relative to other soil water content nodal groupings.

4. Discussion

The dye tracing results for Kongskilde did not reflect notable differences in infiltration depth among the different surface soil water conditions at the site. At the relatively large 'rates' applied ($180,000 \text{ l ha}^{-1}$), the dye reached a depth of at least 60 cm for all run reps except rep 2 for run 8. The high rates,

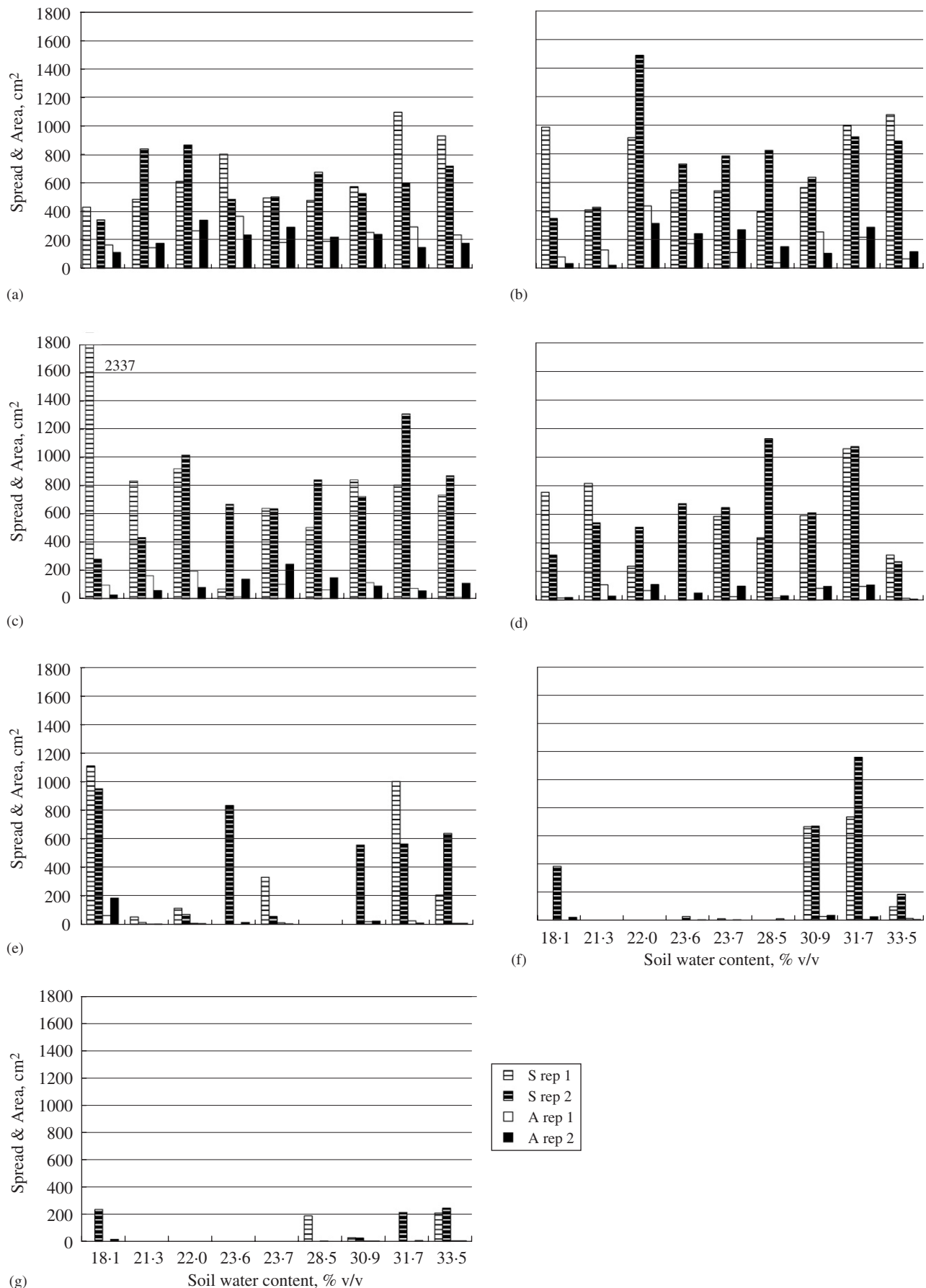


Fig. 7 – Degree of dye spread (S) and area (A) coverage for specific soil depths down to 50 cm for AerWay treatments: (a) 5 cm depth, (b) 10 cm depth, (c) 15 cm depth, (d) 20 cm depth, (e) 30 cm depth, (f) 40 cm depth, (g) 50 cm depth.

Table 4 – Classification and regression tree analysis-based regression tree results for the AerWay treatment. The infiltration variables were considered independent variables (target variables) while soil water content (% v/v) was considered the predictor

Infiltration Variable ^a	Terminal node number ^b	Soil water content group ^c % v/v	Terminal node mean		Term node standard deviation		Number of term. Node observations
			cm	cm ²	cm	cm ²	
D	1	≤ 19.7	67.00		3.00		2
	2	> 19.7 & ≤ 29.7	43.35		9.33		10
	3	> 29.7	81.33		12.53		6
S20	1	≤ 31.3		559.89		259.59	14
	2	> 31.3 & ≤ 32.6		1064.95		8.05	2
	3	> 32.6		293.80		23.10	2
S30	1	≤ 19.7		1027.20		79.30	2
	2	> 19.7 & ≤ 31.3		167.62		256.82	12
	3	> 31.3		600.10		282.16	4
S40	1	≤ 29.7		34.38		104.25	12
	2	> 29.7 & ≤ 32.6		801.53		203.77	4
	3	> 32.6		138.45		44.05	2
A5	1	≤ 21.7		148.18		24.53	4
	2	> 21.7 & ≤ 23.7		297.93		52.15	4
	3	> 23.7 & ≤ 31.3		227.53		36.50	6
	4	> 31.3		211.13		55.75	4
A10	1	≤ 21.7		66.58		42.99	4
	2	> 21.7 & ≤ 22.8		375.05		60.85	2
	3	> 22.8 & ≤ 32.6		185.45		78.31	10
	4	> 32.6		92.00		24.50	2
A20	1	≤ 29.7		48.10		37.55	12
	2	> 29.7 & ≤ 32.6		96.65		8.58	4
	3	> 32.6		10.55		2.95	2
A30	1	≤ 19.7		121.60		60.60	2
	2	> 19.7 & ≤ 29.7		3.72		4.32	10
	3	> 29.7		13.82		7.77	6
A40	1	≤ 19.7		9.25		9.25	2
	2	> 19.7 & ≤ 29.7		1.16		2.45	10
	3	> 29.7		16.30		11.39	6

^a Infiltration variable letters are accompanied with a number reflecting the depth of measurement in cm (e.g., S20 = spread at 20 cm depth). Infiltration variables (depth) not shown in the table did not produce significant cross-validated results (Steinberg & Colla, 1995).

^b Terminal node designation for trees produced with the lowest cross-validation statistic. Maximum terminal node number for each variable indicates total number of terminal nodes in the given CART regression tree (no further tree splitting occurred).

^c Soil water content (% v/v) of given terminal node observations. Bold values = criteria first used to split all available data into two groupings.

which are viable biosolid application rates in Ontario for some environmental and biosolid nutrient situations, were used in lieu of lower rates to assess more fully the effects of furrow side-wall modification on infiltration patterns. Generally, there were no systematic differences in infiltration depth between the different runs perhaps due in part to the narrower observed soil water content range and the large application rates and placement of the material beneath a potentially more sorptive and porous surface soil (i.e., in furrow with furrow bottom at approximately 13 cm depth) that responds more readily to drying and wetting conditions. Moreover, furrow sampling areas were cleared of soil, which would only increase the total volume of infiltrating liquid to

lower soil layers. Nevertheless, tile drainage systems for these soils can exist around 60 cm depth; thus the results suggest that tile or shallow groundwater contamination is likely could occur for this furrow-based approach for the selected application rates irrespective of the soil water content conditions observed.

The soil physical properties of the plough pan (at approximately 20 cm depth) appears significant with respect to modifying infiltration patterns for all water content conditions, as a majority of the dye infiltrated above 20 cm depth (injection depth was simulated at 13 cm). Below roughly the plough pan layer, more discrete macropore flow was evident as observed via the relatively higher dye S:A ratios. Plough

pan effects on solute transport has been observed and modelled by Steenhuis *et al.* (1994) and by Shalit and Steenhuis (1996). Our findings here support their essential experimental and modelling efforts. They noted that applied solutes must first undergo mixing and adsorption in the soil layers above the plough pan. After mixing and saturation in this layer has occurred, solutes flow to depth (sub-pan) in the soil via discrete macropores at much higher flow rates, relative to matrix flow. Lapen *et al.* (2002) noted that the average and standard deviation ($\pm$) of soil cone penetration resistance in the 5–10 and 20–25 cm depth increments at the site were 1.74 ± 0.40 and 3.67 ± 0.83 MPa, respectively. Field-saturated hydraulic conductivity using pressure infiltrometers with rings set into the soil flush with the surface were around 0.03 cm s^{-1} higher than those measured using rings set into the soil at 15 cm depth. The lower hydraulic conductivity of the plough pan is likely linked to soil compaction processes and the collocated fining of texture that occurs between 20 and 30 cm depth. Beneath the pan, the dye spread:area ratios are larger, reflecting primarily spatially constrained infiltration as would be consistent with discrete macropore flow (e.g., worm burrows and root channels). Indeed visual observations of the dye patterns below the plough pan confirm primarily worm burrow transport for most runs. For the driest observed soil conditions, however, there were notably larger numbers of desiccation-cracks and tillage-induced soil fracturing in the top 20 cm which would reduce soil sorptivity (and mixing (Steenhuis *et al.*, 1994)) and allow the infiltrating liquid to become more immediately available to the more continuous macropores beneath the pan (i.e., worm burrows).

In Ontario, the regulations for land application of biosolids require at least 30 cm of unsaturated surface soil. Unsaturated soil is defined as having a water content value less than porosity. In addition, there is no global lower water content limit to application (Nutrient Management Act, 2002). The AerWay infiltration depth results support the conceptual diagram given in Fig. 8; where it is indicated that tile/shallow groundwater-contaminant risk potential is greatest at both

the ‘higher’ and ‘lower’ relative soil water content conditions. The statistical analysis conducted here indicates that, infiltration can occur below 60 cm depth for surface soil water content conditions below approximately 20% v/v and above approximately 30% v/v for these soils. Soil desiccation cracks and greater time-induced soil fracturing potential, as observed especially for the former water content situation, supported flow bypass in the top layers of the soil, making more liquid available for infiltration to deeper depths in the profile via discrete worm-based macropores (Turpin *et al.*, 2007). Such a contention is supported by visual observation of dye flow patterns and the relatively high S:A ratios for the deeper observation depths as per the Kongskilde study. In addition, the lower average dye area extent values for soil depth intervals between 5 and 20 cm for the lower soil water content runs ($\leq 21.7\%$ v/v) could be related to fracture flow bypass to soil depths immediately below the tillage layer (sub-pan). The larger dye area values in the 30 cm depth increment for the run with the lowest average soil water content supports this contention.

Tile drain depths and even shallow groundwater depths could be contaminated by the infiltrating liquid at the simulated application rates used in this study. The AerWay findings, in particular, indicate that for these specific soils, it would be best to apply liquid biosolids to the field for surface soil water contents between approximately 20% v/v and 30% v/v. For situations that might occur in spring, for instance, where significant surface soil drying may occur over wet subsoil conditions, it would be expected that infiltration would indeed penetrate to deeper depths. However, the tillage layer is a first-order soil contact zone (Steenhuis *et al.*, 1994) that would affect infiltration and ultimately infiltration depth to important features like tile drainage networks. And for AerWay results, like the Kongskilde findings, a majority of dye liquid infiltrated to and remained in the soils above the plough pan. Thus, indicators of application timing based on tillage layer soil water content would likely be useful for these conditions as well.

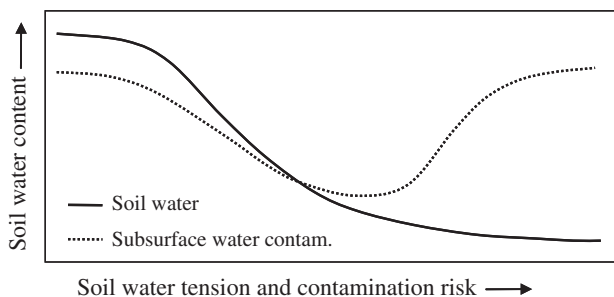


Fig. 8 – Conceptual model of soil water retention and theoretical subsurface (e.g., tile drainage and shallow groundwater) contamination potential for the soils studied. Higher contamination risk at higher water contents associated with reduced sorptivity and larger operative soil pore sizes, and higher risk at lower water contents where soil bypass-based preferential flow is facilitated by desiccation cracks and tillage-induced soil fracturing.

5. Conclusions

This study used Brilliant Blue dye tracers to investigate the nature of infiltration associated with immediate shallow-time action via Kongskilde Vibro-Flex S-tine and AerWay[®] SSD rolling aerator-type tines, for many different soil water conditions of a silty-clay loam soil. One purpose of this work was to help define optimal times, in terms of soil water content status, to apply liquid amendments to land surfaces in order to reduce tile drainage and groundwater contamination potential.

The overall findings can be summarised as follows: For the furrow-based Kongskilde tillage methods, the area and spread of dyed liquid applied to a stretch of furrow was primarily constrained above the plough pan for the entire range of observed furrow-level soil water contents. Effective blue dye liquid application rates were $180,000 \text{ l ha}^{-1}$. Blue dye spread: area ratios were greater below the pan than above it. However, average dye infiltration depths, for all water content conditions, exceeded 60 cm depth. For the AerWay treatment,

which produced discrete soil pockets, was non-linearly related to the surface soil water content; greatest dye penetration depth occurred at the lowest observed soil water contents (<20% v/v) and at the higher ones (>29% v/v). For the lower surface soil water content conditions, desiccation cracks and soil fractures facilitated by tillage action augmented bypass-based preferential flow to continuous sub-pan macropore networks. The greater infiltration depths achieved for the wetter soil conditions were due to reduced sorptivity and greater intrinsic pore sizes that are operative at lower soil tensions. And like the Kongskilde treatment, a majority of the dye remained above the plough pan; and beneath the pan, discrete macropore-based flow (e.g., worm burrows, root channels) dominated infiltration to the deepest observed depths.

Essential findings here underscore the importance of basing liquid amendment timing regulations not only on an upper water content limit but also on a lower one as well. In addition, the study does support the impact of plough pans augmenting sorption in the tillage layer, thus minimising the degree of preferential flow to depth during a typical land application of liquid amendment. Thus, for soils such as these, deeper injection techniques that disrupt the pan and place the liquid amendment beneath it may not be prudent land application options in terms of groundwater and tile drainage contamination.

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